Chapter 4, Problem 44P

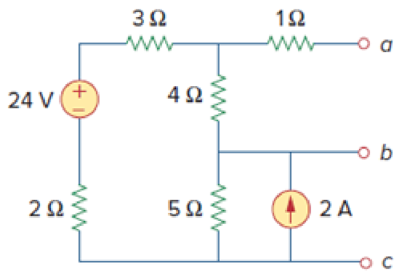
Problem

For the circuit in Fig. 4.111, obtain the Thevenin equivalent as seen from terminals:

(a) *a-b*

(b) *b-c*

Figure. 4.111:



Step-by-step solution

Step 1 of 7

(a)

For R_{Th}, consider the circuit in Fig-(a).

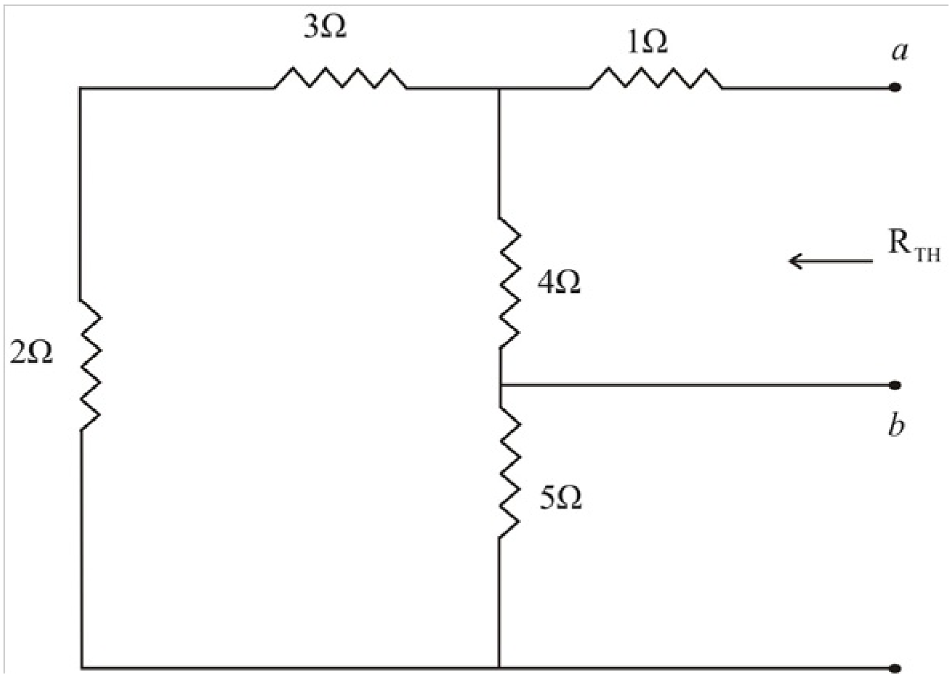


Fig-(a)

Step 2 of 7

$R_{Th} = 1\Omega + (4\Omega \,\|\, (2\Omega + 3\Omega + 5\Omega))$

$R_{Th} = 1 + \frac{4 \times 10}{4 + 10}$

$R_{Th} = 1 + \frac{40}{14}$

$R_{Th} = 1 + 2.857$

$R_{Th} = 3.857\Omega$

Step 3 of 7

For V_{Th}, Consider the circuit in Fig-(b).

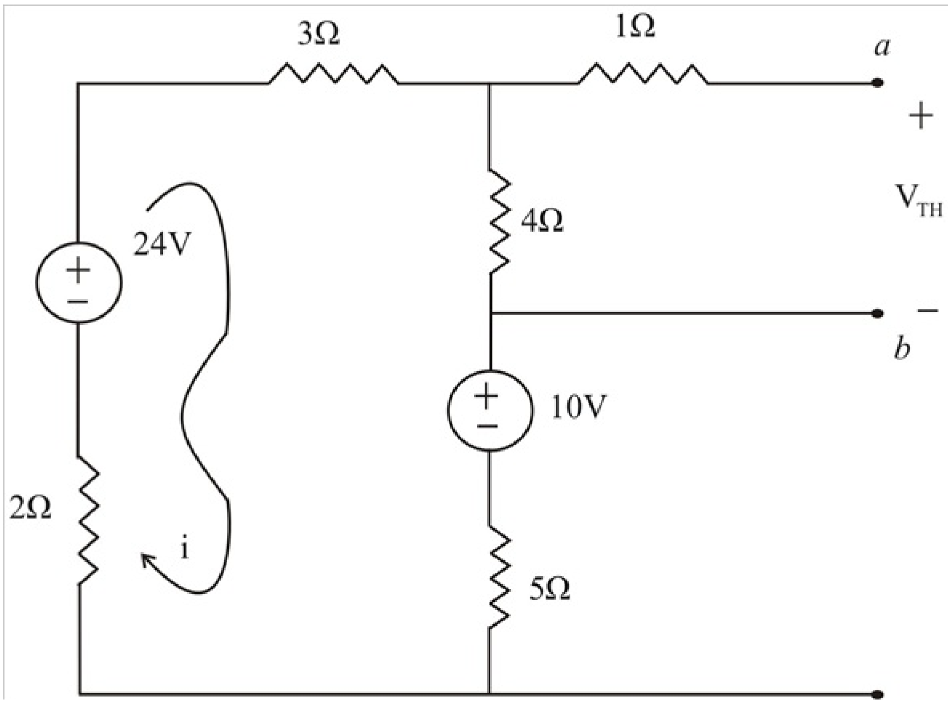


Fig-(b)

Step 4 of 7

Applying KVL gives,

$10 - 24 + i(3 + 4 + 5 + 2) = 0$

$-14 + i(14) = 0$

$i = 1\text{A}$

Therefore,

$V_{Th} = 4 \times i$

$V_{Th} = 4 \times 1$

$V_{Th} = 4\text{V}$

Step 5 of 7

(b)

For R_{Th}, consider the circuit in Fig-(c).

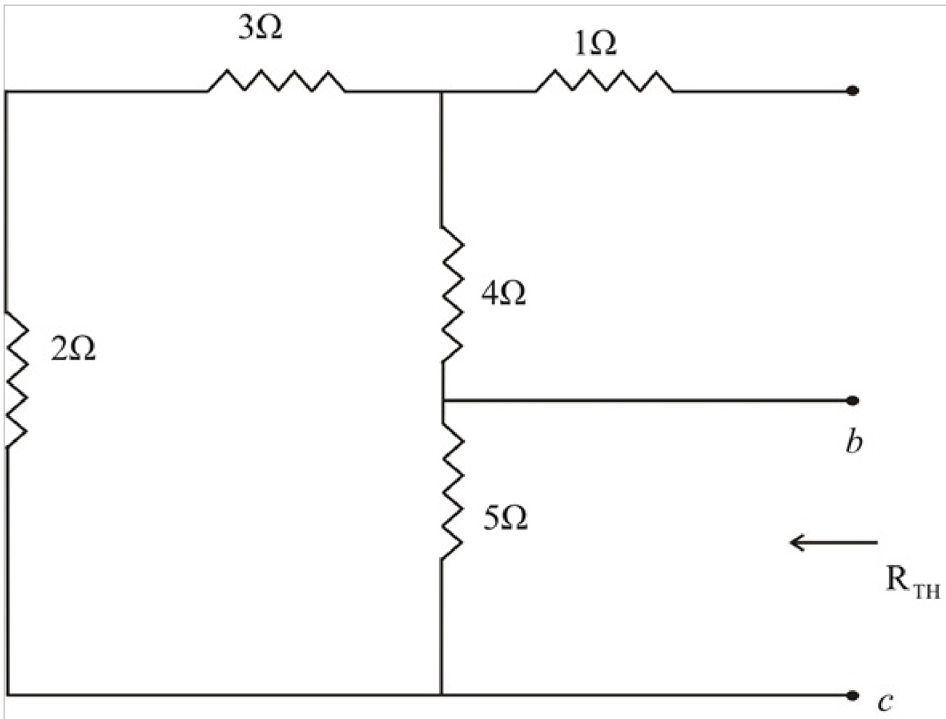


Fig-(c)

Step 6 of 7

$$R_{Th} = 5\Omega \,||\, (2\Omega + 3\Omega + 4\Omega)$$

$$R_{Th} = \frac{5 \times 9}{5 + 9}$$

$$R_{Th} = 3.21\Omega$$

Step 7 of 7

For V_{Th}, Consider the circuit in Fig-(d).

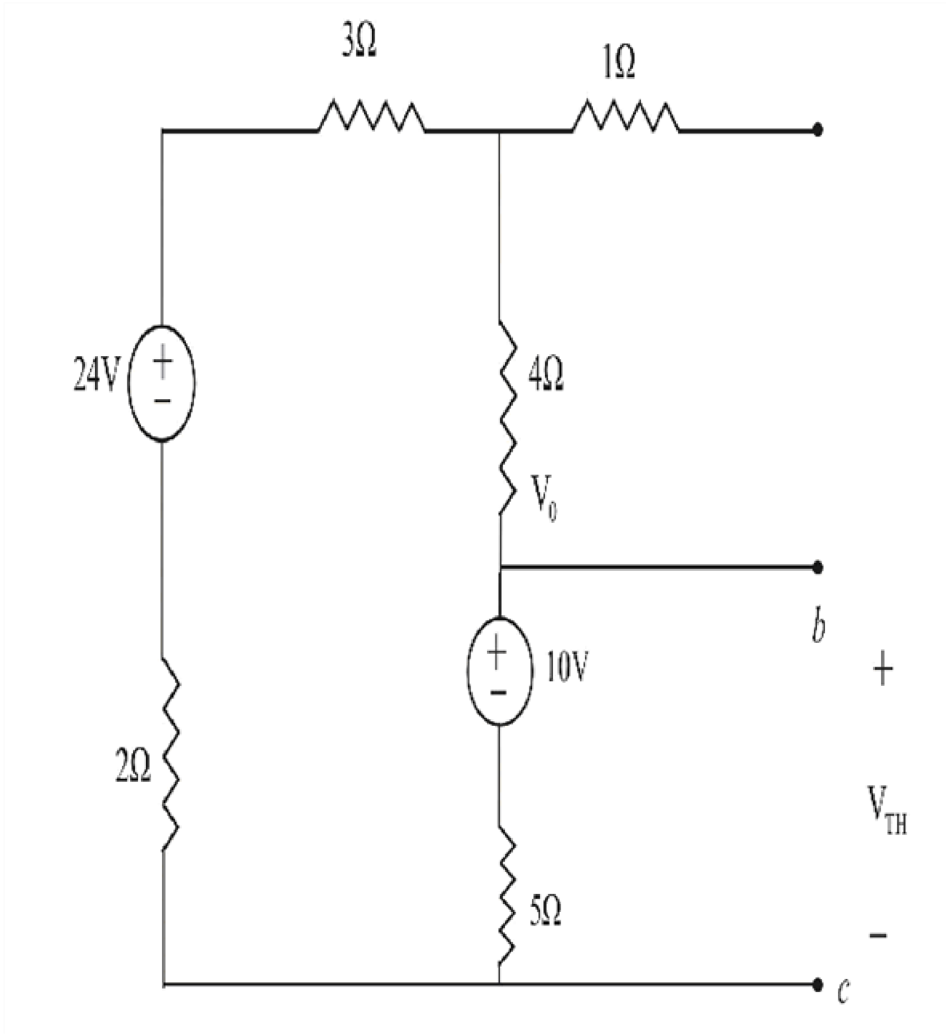


Fig-(d)

Applying KVL gives,

$$10-24+i(3+4+5+2)=0$$

$$-14+i(14)=0$$

$$i=1\text{A}$$

Therefore,

$$V_{Th}=10+5\times i$$

$$V_{Th}=10+5\times 1$$

$$V_{Th}=15\text{V}$$